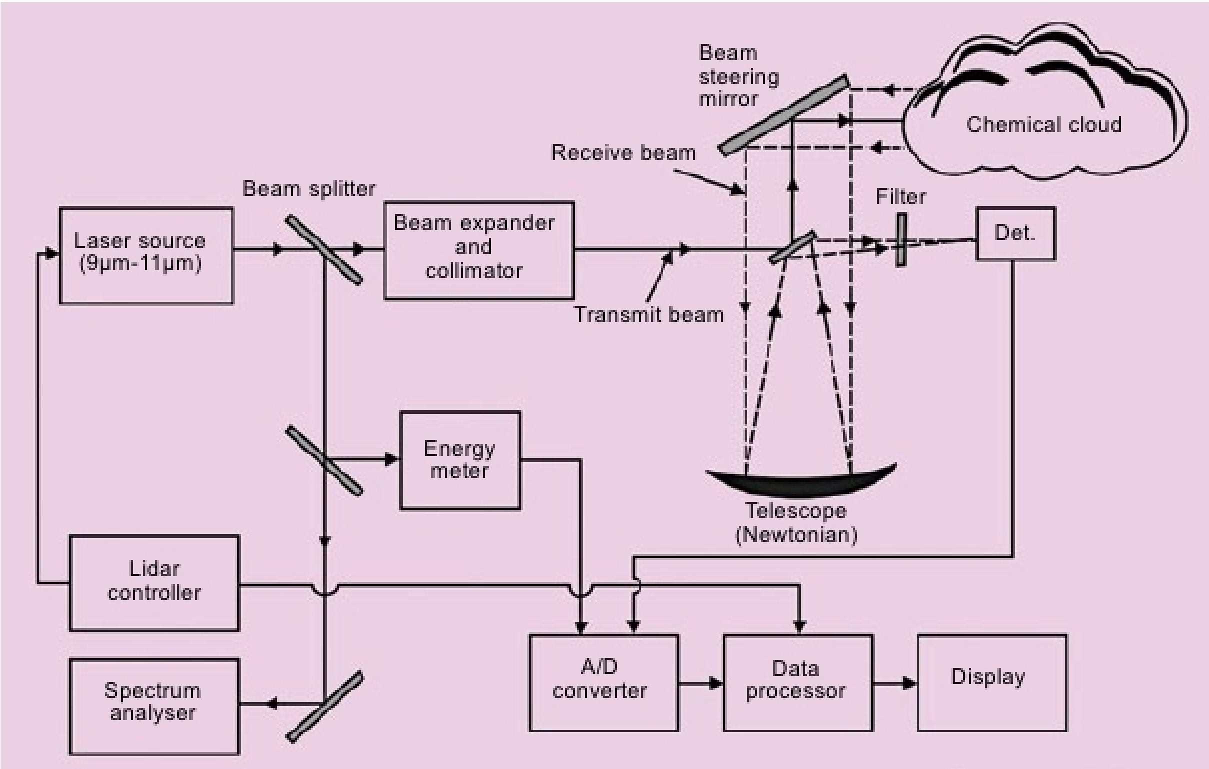

Fig. 3: Differential absorption lidar

Fig. 4: Mobile lidar complex of Laser Systems, Russia

the wavelength of peak absorption of the targeted molecule, and the other to the weak absorption of the targeted molecule. Ratio of the two received backscattered signals measures the concentration of the targeted chemical warfare agent.

Fig. 3 shows the block schematic arrangement of DIAL system. Two laser pulses—called on and off wavelengths—are transmitted towards the targeted area of interest in the atmosphere, which is suspected to be contaminated with a potentially-harmful chemical warfare agent. On wavelength corresponds to the peak absorption of suspected harmful species, while off wavelength is slightly detuned from on wavelength to encounter significantly lower absorption from the same species.

The choice between on and off wavelengths is unique for a given chemical species. It is determined by using a tunable laser that scans the area of interest by transmitting pairs of these wavelengths.

On wavelength, called λ_{ON}, encounters maximum absorption when the corresponding backscattered signal is relatively weak as compared to λ_{OFF}, which encounters weak absorption and produces a relatively stronger backscattered signal. This forms the basis of knowing atmospheric constituents at that time. Ratio of the two backscattered signals gives concentration of the molecules of interest.

Many chemical warfare agents have significant absorption bands in 3µm - 4µm and 9µm - 11µm bands. Carbon dioxide (CO_2) lasers emitting in the latter band have been commonly used for the detection of majority of chemical agents. Some DIAL systems use both these bands and therefore employ both tunable mid-IR as well as CO_2 laser sources. The schematic arrangement of DIAL system shown in Fig. 3, however, makes use of only CO_2 laser.

The transmit laser beam, after suitable collimation, is directed towards the atmosphere in the desired direction with the help of a scanning gimbal mirror. The backscattered signal is received by the receiving telescope and is further focused on to the detector sub-system.

Interference filter is used to block undesired radiation. It allows only the radiation of wavelength of interest to pass through it. Detected signal is then